NGUYEN THANH PHAT

Senior Full-Stack Developer

PHP · Node.js · TypeScript · React

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TECHNICAL SKILLS

Languages
PHP, TypeScript, JavaScript

Backend
PHP/Phalcon, Node.js, NestJS, REST APIs

Frontend
React, Next.js, Tailwind CSS

Databases
PostgreSQL, MongoDB, MariaDB/MySQL, Redis

Messaging
RabbitMQ, Apache Kafka, BullMQ

Identity & Security
Keycloak, SSO, JWT, OAuth 2.0/OIDC, RBAC, SpiceDB/ReBAC

DevOps
Linux, Docker, Docker Compose, Nginx, Apache, Git, GitLab CI/CD, PM2

Testing
Vitest, Jest, Playwright, manual testing

DOMAINS

Network device management
Identity & access management
File storage & sharing
IoT & device integration
Face recognition & attendance

EDUCATION

Engineer's Degree
Computer Networks & Communications
University of Information Technology — VNU-HCM
Graduated 2020

SUMMARY

Senior Full-Stack Developer and hands-on Tech Lead with **10+ years of experience** owning production systems end to end. Strong in backend architecture, system modernization, performance optimization, and production troubleshooting across PHP/Phalcon and Node.js/NestJS stacks. Built and operated a 24/7 platform managing **250,000+ devices**, ~200,000 concurrent connections, and **~5,000 events/minute**. Lead a three-developer team while remaining hands-on across architecture, implementation, deployment, and production support.

EXPERIENCE

HONEYNET Joint Stock Company — 03/2016 – Present

Tech Lead & Full-Stack Developer — 2022 – Present

- Led the full rewrite of a PHP/Phalcon file-storage and identity platform to **NestJS + Next.js**, including architecture and database migration; the new platform serves **5,000+ active users**.
- Designed and built a 24/7 TR-069 platform managing **250,000+ devices**, with ~200,000 concurrently online and **~5,000 events/minute** processed by 20+ RabbitMQ workers.
- Re-architected background processing from a process-per-request model to **RabbitMQ + 20+ workers**, eliminating file-descriptor exhaustion, database connection-pool exhaustion, and CPU/memory overload under production traffic.
- Optimized device queries by limiting retrieved fields and adding targeted indexes, reducing response times from **30+ seconds to a few seconds**.
- Lead three developers while remaining hands-on across architecture, implementation, code review, deployment, troubleshooting and production support.

Full-Stack Developer — 03/2016 – 2021

- Designed and built a Wi-Fi controller from the ground up with PHP/Phalcon and MariaDB for device management, RADIUS, monitoring, reporting and integration APIs.
- Built an internal operations platform covering COVID-19 declarations, QR codes, executive scheduling, task management, Face ID integration and attendance reporting; later upgraded core services from PHP 7 to PHP 8.

SOFT SKILLS

Technical leadership
Lead 3 developers: task breakdown, estimation, technical decisions

Mentoring & coaching
Code quality, debugging, Git, security; stretch tasks with guidance

Planning & delivery
Scope, milestones, progress tracking, re-planning on delays

Ownership & accountability
Technical owner from architecture to 24/7 production support

Problem solving under pressure
Production incidents: log analysis, rollback, performance fixes

Collaborative decision-making
Propose architecture, discuss with the team, adopt better ideas

Cross-functional communication
Work with PM, QA and departments consuming system data

Remote collaboration
Daily updates, Git-based workflow, self-managed deadlines

LANGUAGES

Vietnamese: Native
English: Technical reading & written communication

SELECTED PROJECTS

File Storage, Sharing & Identity Platform
07/2026 – Present

Tech Lead / Full-Stack Developer · NestJS, Next.js, PostgreSQL, SpiceDB, Keycloak, MinIO, Kafka

- Implemented PostgreSQL ltree folder hierarchies, trash, public links and owner/editor/viewer permissions backed by SpiceDB.
- Built a Keycloak administration backend with bulk Excel user import/update, review workflows, audit logging and email tracking/resend.
- Separated HTTP, worker and cron workloads; used Kafka for cross-application events and built an iframe-based File Picker SDK.

ACS Admin — TR-069 Device Management
04/2024 – 09/2026

Tech Lead / Full-Stack Developer · PHP 8.3, Phalcon, MongoDB, GenieACS, RabbitMQ

- Built bulk configuration, diagnostics, network maps, device templates/tags and condition-based alerts via Telegram and email.
- Implemented message identifiers to prevent duplicate processing and used database transactions to maintain consistency across multi-step updates.
- Provided integration APIs used by other departments and systems to consume device data for operational reporting.

Digital Suite for a Ward-Level Government Agency
07 – 09/2026

Tech Lead / Full-Stack Developer · NestJS, TypeORM, Socket.IO, Next.js, Keycloak

- Built an IAM service with bidirectional Keycloak synchronization and a Directory API for other applications.
- Built a paperless meeting system covering schedules, documents, attendees, live voting and controlled impersonation.
- Implemented automated testing across three backend systems using Vitest and Playwright.

Face ID — Access Control & Attendance
07/2022 – 07/2026

Full-Stack Developer · PHP/Phalcon, MongoDB, RabbitMQ, Redis, Firebase

- Built a platform for face-recognition devices, door locks, snapshots and attendance reporting.
- Developed a multi-tenant version with event triggers, alert conditions, recognition schedules and worker-based bulk processing.